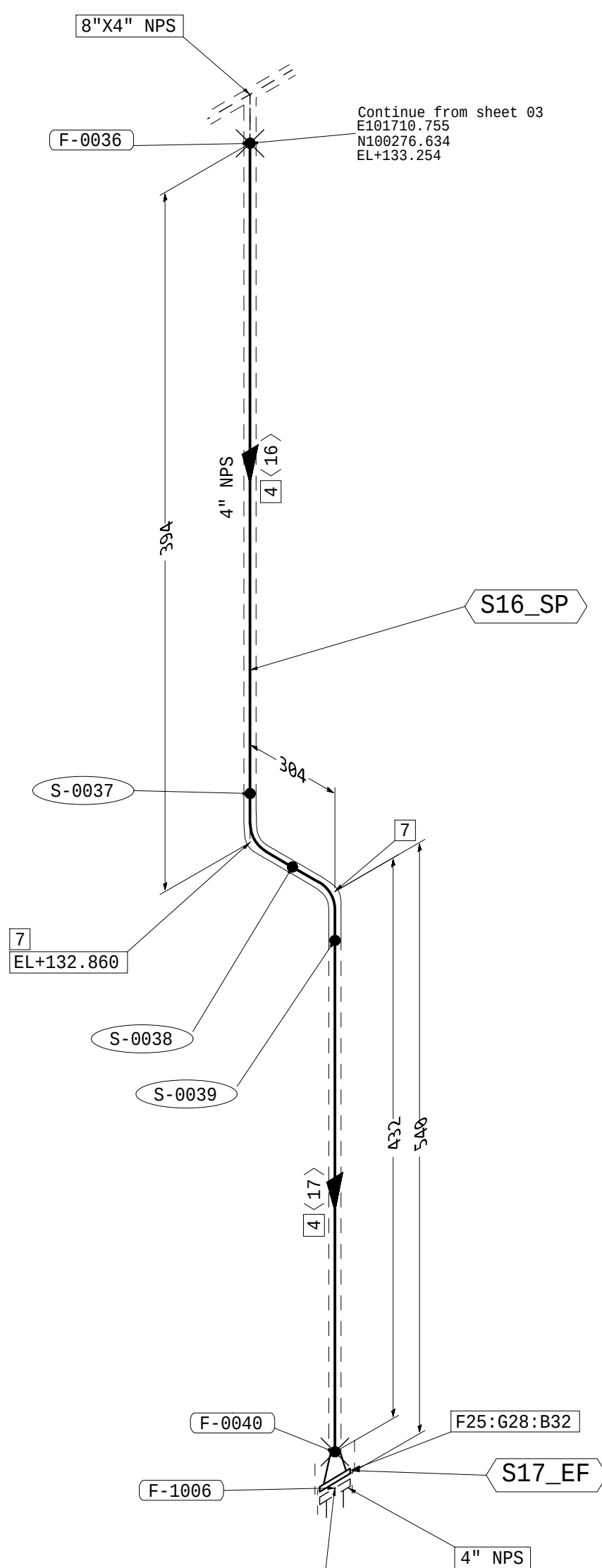
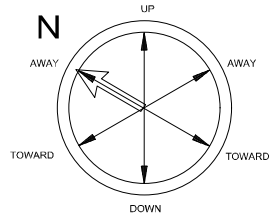


THIS SPPOOL MARKUP DRAWING IS GENERATED FROM ENGINEERING PCF & REFERRING APPROVED ENGG ISOMETRICS. ENGINEERING ISOMETRICS TO BE REFERRED FOR REVISION APPROVAL INFORMATION & IN CASE OF ANY DISCREPANCY DATA GIVEN IN ENGINEERING ISOMETRIC SHALL GOVERN



CONT. TO: 867-E-4205\_03&04/867-E-4205-04\_N1\_A  
4 in RFFE 600  
E101710.755  
N100276.330  
EL+132.317

| MATERIALS                      |                                                                                                  |                |                    |            | STANDARD DETAIL REFERENCE DRAWINGS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| PT No.                         | DESCRIPTION                                                                                      | NPD(IN)        | ITEM CODE          | QTY        | DETAILS HEAVY WELDING BOSSSES FOR THREADED CONNECTIONS<br>CONSTRUCTION DETAILS AND DETAILS FOR ISOMETRIC PIPING<br>INSULATION DETAILS FOR HOT PIPING<br>INSULATION DETAILS FOR COLD PIPING<br>INSTALLATION OF JACK SCREWS FOR FLANGED JOINTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 4                              | PIPE SMLS 4" A106-B SOUR STD BE B36.10M                                                          | 4              | PPS2AERT04ZZABUAIA | 0.522      | AE-036175<br>AD-036077<br>AA-036911<br>AA-036913<br>AC-036930                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 7                              | ELB 90° SMLS LR 4" A234-WPB SOUR STD BE B16.9                                                    | 4              | PE92AZRT04ZZABUAIQ | 2          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 25                             | FLG WNR 4" A105N SOUR STD, 600# 125 B16.5                                                        | 4              | PFY2ADRT04ZZAEMADH | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 28                             | GSK SPR WND W/CS CENTERING RING + I - SS 4" 4.50 MM THICK SS316L+GRPH FILL SOUR 600# B16.20      | 4              | PGUUB5RT04AIAADZCC | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 32                             | BOLT, STUD+2 NUTS A193 GR B7M/ A194 GR 2HM, CERAMIC FLP COAT SOUR B18.2.1/18.2.2 - 160 mm Length | 0.875          | PH8ASCRTAH00160ZZF | 8          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                |                                                                                                  |                |                    |            | <b>NOTES</b><br>1. MATERIAL SHOWN WITH A BROKEN LINE AND SPOOLS LISTED IN THE SAUDI ARAMCO FABRICATION LIST ARE NOT TO BE FURNISHED BY THE VENDOR.<br>2. PIPING DETAILS SHALL BE CUT AT ALL POINTS MARKED F.W. (FIELD WELD).<br>3. EACH FABRICATED SECTION SHALL BE CLEARLY MARKED WITH ITS CODE NUMBER.<br>4. UNLESS OTHERWISE NOTED: (A) ALL DIMENSIONS ARE TO FACE OF FLANGES, (B) BOLT HOLES ON FLANGES STRADDLE C. (C) WELDING ELBOWS ARE LONG RADIUS.<br>5. FABRICATION SHALL BE IN ACCORDANCE WITH SAUDI ARAMCO SPECIFICATION 01-SAMSS-10.<br>6. GASKET ALLOWANCE<br>7. PIPING MATERIAL IS<br><table><thead><tr><th>PIPE SIZE</th><th>SCHEDULE</th><th>SPECIFICATION</th></tr></thead><tbody><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></tbody></table><br>8. WELDING STRESS RELIEF - YES <input type="checkbox"/> NO <input type="checkbox"/> 100% X-RAY <input type="checkbox"/> SPOT X-RAY 01-SAMSS-10 <input type="checkbox"/><br>9. TESTING SHOP HYDROSTATIC TEST - YES <input type="checkbox"/> NO <input type="checkbox"/><br>THE SHOP HYDROSTATIC TEST SHALL BE CARRIED OUT ON PIECES HAVING ALL ENDS FLANGED, HYDROSTATIC TEST PRESSURE:<br>10. ALL PIPING MATERIAL IS IN ACCORDANCE WITH SAUDI ARAMCO ENGINEERING SPECIFICATION SAES-L-105 CLASS . ALL EXCEPTIONS ARE NOTED ON THE ISOMETRIC.<br>11. INSULATION REQUIRED - YES <input type="checkbox"/> NO <input type="checkbox"/><br>INSULATION SHOWN ON DRAWING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  | PIPE SIZE   | SCHEDULE | SPECIFICATION |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PIPE SIZE                      | SCHEDULE                                                                                         | SPECIFICATION  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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|                                |                                                                                                  |                |                    |            | <b>SPOOLS TO BE FABRICATED BY VENDOR</b> <table><thead><tr><th>NO. PIECES</th><th>MARK NO.</th><th>SIZE</th><th>WEIGHT LBS</th><th>NO. PIECES</th><th>MARK NO.</th><th>SIZE</th><th>WEIGHT LBS</th></tr></thead><tbody><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  | NO. PIECES  | MARK NO. | SIZE          | WEIGHT LBS | NO. PIECES | MARK NO.      | SIZE                   | WEIGHT LBS |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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|                                |                                                                                                  |                |                    |            | <b>SPOOLS TO BE FABRICATED BY SAUDI ARAMCO</b> <table><thead><tr><th>NO. PIECES</th><th>MARK NO.</th><th>SIZE</th><th>NO. PIECES</th><th>MARK NO.</th><th>SIZE</th></tr></thead><tbody><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  | NO. PIECES  | MARK NO. | SIZE          | NO. PIECES | MARK NO.   | SIZE          |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| NO. PIECES                     | MARK NO.                                                                                         | SIZE           | NO. PIECES         | MARK NO.   | SIZE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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|                                |                                                                                                  |                |                    |            | <b>REFERENCE DRAWINGS</b> <table><thead><tr><th colspan="2">PIPING PLAN</th><th>YA-212287</th></tr><tr><th colspan="2">P&amp;ID NO.</th><th>YA-221177-001</th></tr><tr><th colspan="2">PLOT PLAN KEY PLAN NO.</th><th>YA-212083</th></tr><tr><th colspan="2">INDEX FOR PIPING ISO (AG)</th><th>PP-867-LI-0003</th></tr><tr><th colspan="2">INDEX FOR PIPING ISO (UG)</th><th>PP-867-LI-0004</th></tr></thead></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  | PIPING PLAN |          | YA-212287     | P&ID NO.   |            | YA-221177-001 | PLOT PLAN KEY PLAN NO. |            | YA-212083 | INDEX FOR PIPING ISO (AG) |  | PP-867-LI-0003 | INDEX FOR PIPING ISO (UG) |  | PP-867-LI-0004 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PIPING PLAN                    |                                                                                                  | YA-212287      |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| P&ID NO.                       |                                                                                                  | YA-221177-001  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PLOT PLAN KEY PLAN NO.         |                                                                                                  | YA-212083      |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| INDEX FOR PIPING ISO (AG)      |                                                                                                  | PP-867-LI-0003 |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| INDEX FOR PIPING ISO (UG)      |                                                                                                  | PP-867-LI-0004 |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>PIPE CUT LENGTH SUMMARY</b> |                                                                                                  |                |                    |            | SXX_Spool Sequence Number (S01,02, SXX_SP - Fab. Spool<br>SXX_PP - Fab. P to P Doubling (0-6M+0M)<br>SXX_SP - Erec. Random Pipe (0M,12M)<br>SXX_EP - Erec. Part Loose Fittings<br>SXX_EV - Erec. valve<br>SXX_EPG - Erec. Pipe Plug<br>SXX_SBP - Small Bore Spool Fab.<br>SXX_SBE - Small Bore Spool Erec.<br>SXX_SBD - Small Bore Spool Drain<br>SXX_SBV - Small Bore Spool Vent<br><input checked="" type="checkbox"/> PART NUMBER<br>-XX- PIECE NUMBER<br>-XX- SHOP WELD<br>-XX- FIELD WELD<br>-XX-01- SPOOL NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SPL No.                        | NPD                                                                                              | PC No.         | LENGTH             | ALLOWANCE  | TOTAL LENGTH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ISO For                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| S16_SP                         | 4                                                                                                | <16>           | 0.242              | 0.000      | 0.242                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | NOTES :<br>1. ALL COORDINATES AND ELEVATIONS ARE IN 'METER' AND ALL DIMENSIONS ARE IN 'MM'.<br>2. PIPE QUANTITY IS IN METERS AND OTHER QUANTITY IN NUMBERS.<br>3. FOR STANDARD SUPPORT, REFER DOC. NO. "PP-868-ST-0001".<br>4. IN CASE OF ANY DISCREPANCY FOR LINE CONDITIONS, LINE DATA GIVEN IN LINE LIST SHALL GOVERN<br>5. FOR SMALL BORE LINES, DIMENSION & ROUTING SHALL BE FIELD ADJUSTED.<br>6. THE SPECIFIED THEORETICAL LOCATION OF EQUIPMENT NOZZLES, SUPPORTING STRUCTURE SHALL BE VERIFIED BEFORE INSTALLING THE PIPING & INTOLERABLE DISCREPANCIES SHALL BE DISCUSSED WITH DESIGN SITE ENGINEER.<br>7. ALL THREADED CONNECTION SHALL BE SEAL WELDED, EXCEPT DRAIN AND VENT PLUG.<br>8. REFER LINE LIST.<br>9. ALL PRESSURE WELDS, OTHER THAN BUTT WELDS, INCLUDING BRANCH CONNECTIONS SUCH AS BRANCH-OLETS, AND BOSS) SHALL BE 100% PT/MT EXAMINED. ALL ATTACHMENT WELDS ON PRESSURIZED COMPONENTS THAT ARE NOT HYDROTESTED AFTER MAKING THE ATTACHMENT WELD SHALL BE 100% PT/MT EXAMINED. |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| S16_SP                         | 4                                                                                                | <17>           | 0.280              | 0.000      | 0.280                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | BUTT WELD<br>FIELD WELD<br>INSULATED PIPE<br>TRACED PIPE<br>PIPE SUPPORT<br>PLATFORM PENETRATION<br>SOCKET WELD<br>SCREWED JOINT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                |                                                                                                  |                |                    |            | DESIGN CERTIFICATION<br>(DESIGN AGENCY)<br>ENGG. DATE<br>CERTIFIED BY DATE<br>CONTR. AGENCY DATE<br>OPERATIONS DATE<br>BY DATE<br>REV. NO.<br>DATE<br>REVISION DESCRIPTION<br>JOB ORDER/REV. NO.<br>10-14815-0003<br>THIS REVISION IS NOT TO BE USED FOR CONSTRUCTION (UNLESS CERTIFIED AND DATED)<br><b>SAUDI ARABIAN OIL COMPANY</b><br>ISOM 8" - MDEA-17357-1CH2P06<br>MDEA SERVICE<br>JFGP GAS PROCESSING TRAIN-4<br>JAFURAH<br>SAUDI ARABIA<br>PLANT NO.<br>INDEX<br>DRAWING NUMBER<br>SHT. NO.<br>REV. NO.<br>867<br>L<br>YB-215304<br>050f05<br>00A<br>SITE REV. NO.<br>00A<br>RESTRICTED<br>SMART<br>CADD-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                |                                                                                                  |                |                    |            | NOTES CONT. :<br>91. FOR SOUR SERVICE PWHT IS MANDATORY FOR 1. DOUBLE SIDED WELDING, 2. INTERNAL WELD REPAIR 3. INTERNAL WELD BUILD UP FOR DISSIMILAR THICKNESS JOINTS TO OVERCOME MISALIGNMENT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                |                                                                                                  |                |                    |            | PWHT<br>CHEMICAL CLEANING<br>PIPING MATERIAL CLASS<br>DESIGN<br>TEMP. (DEG. F)<br>PRESS. (PSIG)<br>OPERATING<br>TEMP. (DEG.F)<br>PRESS. (PSIG)<br>TEST<br>PRESS. (PSIG)<br>TEST MEDIUM<br>RADIOGRAPHY %<br>SHOP<br>FIELD<br>INSULATION<br>THK. (MM)<br>TYPE<br>TRACING<br>PAINTING<br>STRESS SYS NO.<br>LINE ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                |                                                                                                  |                |                    |            | Y<br>NOTE-8<br>1CH2P06<br>270<br>170<br>172<br>100<br>428<br>WATER<br>10<br>10<br>25<br>HC<br>NA<br>APCS-2A<br>867-GPT4-154<br>MDEA-17357-420A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |  |  |             |          |               |            |            |               |                        |            |           |                           |  |                |                           |  |                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

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